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Research article

A multi-location inventory system with occasional allocation under uncertain defective rates

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ABSTRACT

In this study, a multi-location allocation problem is investigated under a temporary allocation system in light of the uncertain defective rate inherent to items. Based on the theoretical and modeling foundation of previous research, this study contributes in four main aspects. First, a robust optimization model is developed based on the budget uncertainty of the defective rate of items in a multi-location inventory system. Second, given that the determination of uncertainty relies on the subjective judgment of decision-makers with different risk preferences, a pivotal variable method is introduced to approximate the possible thresholds of the uncertain parameter within the expected interval. Third, a new evaluation criterion curve, illustrating the cumulative cost differences between the probabilistic and robust models, is employed through large-scale sampling to evaluate the performance of the two decision models. Finally, numerical experiments and sensitivity analysis are conducted to verify the performance of the decision effects under different defective rates and stockout costs, respectively. The results confirm the superior performance of the robust model, as shown through sensitivity analysis under thresholds of varying defective rates and stockout costs, and highlight the effectiveness of the proposed cumulative cost difference evaluation curve. The robust model offers significant cost-control advantages over the probabilistic model, particularly in high-uncertainty scenarios. This study provides a reference method for multi-location inventory allocation decisions with uncertain defective rates.

1. Introduction

In multi-location inventory systems, the replenished items at retailers' stores based on initial orders often produce deviations from the original order, even under the condition of deterministic customer demand. This phenomenon is particularly pronounced because of the overlap of production processes and logistics operations within a compressed service time, which leads to the frequent identification of unsuitable/defective items at retail terminals. With the prevalence of individualization requirements, small-batch or make-to-order (MTO) production has become the main method. Consequently, when there is a stockout of one type of item, there may be no extra backup storage in the warehouse. The only way is to deliver an alternative from other retail stores in a timely manner to guarantee order fulfillment. As the available items in the market are limited, traditional defectiveness-based stockout optimization based on the assumptions of the post-production defective rates (Dhahri et al., 2022), sales statistical inference defective rates (Cui et al., 2020; Wu and Ouyang, 2001), and experience-based fixed proportion defective rates (Dehghani and Abbasi, 2018) are not suitable for the response of sparse allocation orders. Therefore, a proper prediction of the defective rate uncertainty in a deterministic multi-location inventory environment with occasional allocation requirements becomes a foremost problem in practice and research.

In practice, multi-location inventory allocation deals with the unmet requirements from the first round of sales, which provides essential supplement responses for the perfect fulfillment of orders for individual locations. However, how to make a balance decision over the whole network is still a great challenge for managers (Van Wijk et al., 2012), especially under the condition of items with uncertain defective rates and occasional allocation requirements. Compared to previous studies addressing allocation problems considering defective items following a specific probability distribution, the issue under the condition that the

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order size is already known but the distribution features of defective rates of a kind of item are unclear has not received sufficient research attention. In summary, a method is required to address the uncertain defective rates of items that neither rely on simple proportion settings nor on exact parameter distribution information.

In recent years, robust optimization has gained increasing attention from researchers and decision makers owing to its independence from the distribution characteristics of uncertain parameters; however, it requires only that these parameters satisfy certain properties, such as box, ellipsoidal, and budget uncertainty sets (Ben-Tal et al., 2013). Satisfactory optimization outcomes have been achieved in various decision-making fields, including inventory management (Zhuang et al., 2022), path optimization (Zhang et al., 2021), and resource allocation (Bertsimas et al., 2011). In general, robust optimization is particularly suitable for handling multi-location allocation problems under defective rate uncertainty for two main reasons. First, it replaces the uncertain distributions of variables with data-based modeling, thereby addressing the challenge of fitting probability distributions of uncertain variables in different decision scenarios. Second, robust optimization aims to achieve the best possible outcome without violating constraints under the most adverse conditions. The robustness of the robust optimization results makes them particularly suitable for dealing with issues of additional defect detection during and after the allocation decision. However, employing robust optimization in modeling presents two significant challenges: On the one hand, the construction and solution of the multi-location allocation model based on robust optimization are interrelated, and this is of great importance for the successful transformation of subsequent robust counterparts based on the proposed model structure. On the other hand, verifying the decision performance of the proposed model is a challenging endeavor, necessitating a redesign of the performance evaluation criterion to specifically assess the comparative advantages of the decision results.

Based on this background, this study presents a novel multi-location inventory allocation model for uncertain defective rates. The uncertain parameter follows the characteristics of parameter budget uncertainty sets. The main innovations and contributions of this study are as follows. (1) A novel multi-location inventory model is constructed, and occasional allocation is induced owing to the uncertain defective rate. (2) A pivotal variable method is used to evaluate the deviation of the budget uncertainty parameter. Given that the adjustment of the conservativeness of budget uncertainty parameters is based on the decision makers' subjective risk preferences, this study introduces a pivotal variable method to provide an approximate probability of uncertain variables falling within the expected range, offering a basis for determining the perturbation range of uncertain variables. (3) A performance evaluation curve is developed based on the cumulative reduced cost between the results of the sampling-based decision and the robust decision. To verify the performance of the robust model over the traditional probabilistic decision model, this study designs a new evaluation criterion, which is formed by the cumulative cost differences (or reduced cost of the two models) of the probabilistic and robust models under different parameter settings. The characteristics of the cumulative cost differences are applied based on the designed comparison procedure to evaluate the decision performance of the proposed model.

The remainder of this paper is organized as follows. Section 2 conducts a literature reviews the state-of-the-art research on multi-location allocations. Section 3 introduces the problem and presents the uncertainty model and analyzes the features of the proposed uncertainty model. Section 4 presents the proposed performance verification process and procedure. Numerical experiments are presented in Section 5, and the computational results are collected and analyzed. Section 6 provides the theoretical and managerial implications. Finally, Section 7 concludes the paper and provides directions for future research.

2. Literature review

As the problem of optimal replenishment and inventory allocation for a two-location inventory system was proposed (Gross, 1963), the issue of two- or multi-location (production, inventory, or sales) allocation has garnered significant academic attention (Xu et al., 2024). The horizontal allocation of resources or items from certain locations after the demand has been partially fulfilled plays a critical role in managing out-of-stocks and balancing inventory in multi-location inventory systems (Nonås and Jörnsten, 2007). Effective horizontal allocation decisions are not only helpful in sustaining high-quality service levels and reducing system costs but also have a profound impact on the sustainability of replenishment and service differentiation among multi-location (Olsson, 2009).

The allocation problem under uncertain conditions has garnered significant academic attention (Achamrah et al., 2022), whereas research on uncertain defective rates is relatively scarce. For instance, in emergency situations, multiple uncertainties, such as uncertain demand, supply, and transportation time, usually have accumulated impacts on final decisions (Zhang et al., 2025). Timajchi et al. (2019) studied the inventory allocation and transportation problems of hazardous products under a given deterioration rate and accidental risks. For a certain proportion of quality-unreliable parts, Samuel et al. (2021) established a multicomponent multiproduct transshipment model for part repair and replacement in a multi-location replenishment allocation system. In general, for some cold-chain-transported items, such as blood products, pharmaceuticals, and chemical raw materials, allocation plans are often pre-arranged based on empirical assumptions over the entire logistics process to reduce losses and risks to the largest extent (Dehghani and Abbasi, 2018). However, owing to the overlapping of multiple operations in the areas of production, transportation, and storage (Konstantaras et al., 2012), unexpected damages, losses, and wastage frequently occur, resulting in statistical or empirical decisions that cannot respond to the just-in-time discrete delivery requirements (Dehghani et al., 2021).

Considering item defectiveness, decisions on replenishment and allocation have been addressed by numerous existing studies. Many studies have assumed either a fixed defective rate or a defective rate that conforms to a specific probability distribution. For example, Ongkunaruk et al. (2016) investigated a joint replenishment problem with shipment constraints under empirically predetermined defective rates. Lee and Kim (2014) developed a single-vendor single-buyer integrated production–distribution model, assuming that the defective rates follow a uniform distribution. However, such empirical assumptions oversimplify distributional hypotheses and neglect the diversity of defective rate patterns, thereby constraining the applicability of these models to various scenarios, such as manufacturing stages (Tayyab and Sarkar, 2016) and multi-location inventory replenishment systems (Wang et al., 2023). In general, differentiation in inventory allocation model design is primarily reflected in the diversity of allocation processes, network echelons, and handling logic for defective items.

In production and allocation systems, defective items are typically identified and removed during quality inspections. However, rework processes may lead to delays in allocation, necessitating the simultaneous consideration of production scheduling and transfer timeliness. Gautam et al. (2022) proposed a dual-path management framework for processing defective items at production sites; non-reworkable defects are discounted during allocation, whereas reworkable items are reintegrated into production cycles. In contrast, some logistics systems tend to distribute defect-related risks to receivers by transferring accountability to the transportation or warehousing phases (Chen et al., 2019). Furthermore, the complexity of a logistics network critically depends on the model architecture. Single-echelon allocation systems generally assume centralized defect disposal at the origin nodes, whereas multi-echelon systems require tracking defects and inducing allocation

across different locations (Das et al., 2017). For instance, Wang et al. (2022) developed a stochastic integrated location-inventory-routing model for multi-echelon supply chains by employing hybrid particle swarm optimization and adaptive differential evolution algorithms to synchronize retailer allocation with safety stock optimization. In summary, these studies highlight the necessity of constructing scenario-specific allocation models under defective rate uncertainty.

Conventional studies have constructed numerous uncertainty allocation models by presuming known statistical distributions of defects, although they challenge decision-makers in acquiring distributional information for uncertain parameters under practical scenarios. To overcome this limitation, a robust optimization approach has been developed that characterizes or describes uncertain parameters through uncertainty sets such as box (Soyster, 1973), ellipsoidal (Ben-Tal and Nemirovski, 1999), and budget uncertainty sets (Bertsimas and Sim, 2004), rather than relying on explicit probabilistic distributions. However, robust models that address specific allocation problems with complex objectives and constraints are inherently nonlinear. This complexity poses significant computational challenges, making it difficult to efficiently solve these models. To cope with these challenges, common approaches involve introducing additional constraints within uncertainty sets and leveraging the duality theory to reformulate the original model into a linear counterpart (Ghamari et al., 2024) or employing the Benders decomposition algorithm (Li et al., 2024) to reduce computational complexity.

To achieve a balance between decision conservatism and allocation flexibility, an uncertainty set with distinct geometric and operational properties is an appropriate choice that matches the characteristics of the uncertainty parameter and ensures the feasibility of transshipment strategies under worst-case scenarios. Compared to the budget uncertainty set, the ellipsoidal uncertainty set addresses robust optimization problems through the second-order cone constraints, yet these cones cause nonlinearities that increase computational complexity, the polyhedral uncertainty set, benefiting from linear structures for tractable solutions and easier uncertainty magnitude control though, fails to capture correlations among uncertain parameters and may yield overly conservative solutions, whereas the budget uncertainty set specifies two distinct benefits: (1) ease of transformation of the original model into a computationally tractable robust linear formulation, and (2) providing adjustable solution conservativeness through the defective rate uncertainty budget parameter, which helps enable more customized risk management strategies (Zhang et al., 2025).

The characteristics of the related papers on the construction of uncertain allocation models are listed in Table 1. In previous multi-location inventory allocation problems, the defective rate often exhibited high uncertainty; however, the assumptions of the defective rates are identified following specific distributions or are empirically prescribed. In contrast to previous studies, this study simultaneously incorporates defective rate considerations within robust frameworks and remains relatively scarce.

3. Model formulation

3.1. Problem description

In a system involving multiple retail outlets, such as branded apparel network distribution, multi-location production with maintenance spare parts requirements, and automotive sales networks, supply contracts often comprise a specific quantity of items/products, incorporating the existence of a certain number of defective products. Retailers make inventory decisions based on the execution of contracted orders. However, due to discrepancies between actual demand and the estimated number of order quantities of items with a certain proportion of defective items, it is common for some retail outlets to experience stockouts, while others face overstocks. To fulfill these deterministic real-time orders as quickly as possible, there is a constant flow of transfers or allocations from one location to another, resulting in significant waste.

Under such circumstances, this study considers an inventory allocation system comprising a core platform in charge of scheduling and multiple retailers receiving items. The scheduling platform is responsible for integrated inventory decisions across all retail outlets, as illustrated in Fig. 1. Due to capacity limitations and the presence of defective items, retail outlets often order a fixed number of items and require secondary allocation if temporary demand surpasses storage. In general, the order size for such high-priced items is small, and a temporary unmet demand is not common practice; thus, it challenges the traditional statistical assumption-based stochastic methods to process these discrete distributed orders. Moreover, because the number of items in the whole system is limited, the uncertainty of defective rates of an item is becoming a critical factor that cannot be neglected in allocation practices: (1) The dispatcher faces the risk of stockouts due to the existence of defective items. (2) Defective items in the allocated batches, meaning that the recipient may still be unable to meet their demands after receiving the reallocated items. It is difficult to accurately estimate the number of defective items using experience-based methods or statistical distribution probability methods to construct a robust model for processing these small demands. In summary, this study investigates the total cost minimization problem of a multi-location inventory allocation system under defective rate uncertainty (uncertain multi-location item allocation, UMLIA).

3.2. Assumptions and notes

This study investigates a limited number of items in the system, where each location holds only one or a small share of items, the allocation order is small and sparse, and a shortage can be satisfied by other very limited locations. Based on the characteristics of the problem mentioned in this study, the following assumptions are made. (1) The central platform makes the inventory allocation decision, and there is no price competition for allocation among retailers. (2) Different retailers are located in different regions; thus, the demand of each retailer does not

Table 1
Characteristics of different research models.

Literature	Type of uncertainty	Characteristic of uncertainty	Optimized tools & methods	Robust optimization	Flexibility	Computational complexity
Ongkunaruk et al. (2016)	Defective rate	A given value	Genetic algorithm	–	Low	Low
Lee and Kim (2014)	Defective rate	Uniform distribution	Analytical method	–	Moderate	Moderate
Tayyab and Sarkar (2016)	Defective rate	Beta distribution	Mixed integer programming	–	Moderate	Moderate
Qiu et al. (2017)	Defective rate and demand	Box and ellipsoidal set	Dynamic programming	✓	High	Moderate
Ghamari et al. (2024)	Demand	Budget uncertainty set	Multi-objective gray wolf algorithm	✓	High	Low
Li et al. (2024)	Demand	Budget uncertainty set	Benders decomposition algorithm	✓	High	High
Current study	Defective rate	Budget uncertainty set	CPLEX	✓	High	Low

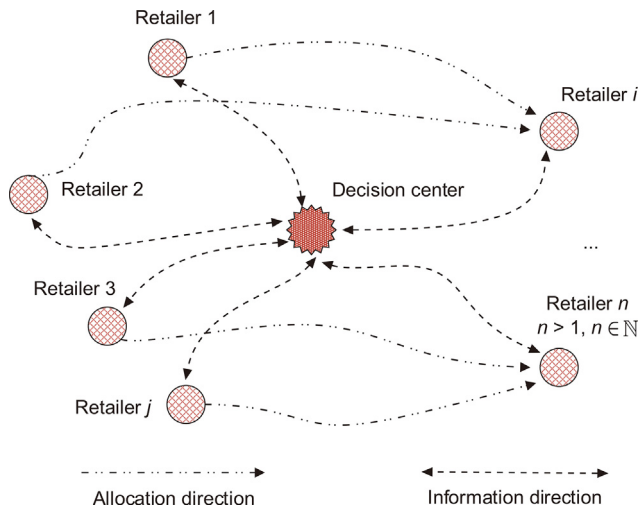

Fig. 1. Network architecture of the multi-location allocation system.

interfere with that of others. (3) Each retailer dispatches items to only one retailer (to guarantee its own order fulfillment), but the retailer can receive the allocated orders from multiple retailers. This guarantees the elimination of invalid circles. (4) Defective items are identified as soon as customers received all the items. (5) Customers with defective items can only exchange information with the retailers selling these items. Therefore, if a retailer has excess items after allocation, it can still provide exchange services within its region. Otherwise, it results in a stockout. (6) The handling of defective items is not part of the allocation system; therefore, handling costs are not considered. (7) Reverse logistics of defective items after allocation is not considered in this study. The model parameters used in this study based on these assumptions are listed in Table 2.

3.3. The UMLIA model

Based on the above description, the proposed UMLIA uncertainty model is presented in Eqs. (1)–(11):

$$minf = \sum_{i\in N} c_h * \left(k_i - d_i - \sum_{j\in N} l_{ij}\right)^+ + \sum_{i\in N}\sum_{j\in N} c_l * l_{ij}$$

$$+\sum_{j\in N} c_b * \left(d_j - \omega_j^s * k_j - \sum_{i\in N} \omega_{ij}^l * l_{ij}\right)^+ \tag{1}$$

$$\sum_{i\in N} \omega_{ij}^l * l_{ij} \le \left(d_j - \omega_j^s * k_j + h_j\right)^+, \forall i \ne j, j \in N \tag{2}$$

$$\sum_{i\in N} l_{ij} + k_j \le s_j, \forall i \ne j, j \in N \tag{3}$$

$$\sum_{j\in N} l_{ij} \le \left(\omega_i^s * k_i - d_i - h_i\right)^+, \forall i \ne j, i \in N \tag{4}$$

$$\omega_{ij}^l = 1 - \tau_{ij}^l \tag{5}$$

$$\omega_i^s = 1 - \tau_i^s \tag{6}$$

$$x_{ij} = \begin{cases} 1 & l_{ij} > 0 \\ 0 & l_{ij} = 0 \end{cases}, \forall i \ne j, i, j \in N \tag{7}$$

$$\sum_{i\in N} x_{ij} \le n - 1, \forall i \ne j, j \in N \tag{8}$$

$$\sum_{j\in N} x_{ij} \le 1, \forall i \ne j, i \in N \tag{9}$$

$$0 \le x_{ij} + x_{ji} \le 1, \forall i \ne j, i, j \in N \tag{10}$$

$$s_i, k_i, d_i, l_{ij} \ge 0, \forall i, j \in N \tag{11}$$

where Eq. (1) is the objective function representing the minimization of the total system cost, which consists of three components: inventory holding cost, allocation cost, and stockout costs. The notation $(a)^+$ denotes $max\ \{a, 0\}$, and the following expressions hold the same meaning as $(a)^+$. Eqs. (2)–(10) are the constraints of the model, and the meaning of each equation is as that: Eq. (2) specifies that the inventory level of retailer j after receiving the allocated items from other retailers, must not exceed its safety stock level. When formulating this constraint from the out-of-stock retailer's perspective, it is essential to evaluate the existence of defective items in both their own inventory and the items allocated by supplier retailers. Eq. (3) ensures that the remaining quantity of items for retailer j, after receiving allocations from other retailers cannot exceed retailer j's maximum inventory capacity. This constraint takes effect from the perspective of the retailer with stockouts; thus, defective items should be considered in the holding inventory and allocated items. Eq. (4) ensures that retailer i, after dispatching allocations, retailer i maintains that the remaining quantity of items is not less than its safety stock level. This constraint takes effect from the perspective of dispatching retailers based on the premise that the items remaining after allocation should meet their own demands. Eqs. (5) and (6) describe the transition relationship between defective and yield rates. Eq. (7) determines whether retailer i will dispatch items to retailer j. Eq. (8) allows retailer j to receive allocations from other retailers. Eq. (9) indicates that considering inventory allocation as an occasional temporary activity requiring additional resources, retailer i can dispatch allocations to at most one retailer, which is different from the function of regional distribution centers with free allocation destinations. Eq. (10) determines the allocation rule that only a retailer with a stock-out accepts allocations from other retailers, and the retailer receiving the allocation will not allocate items to the third retailer. Eq. (11) ensures that all relevant parameters are non-negative.

Table 2
Parameters and definitions.

Notation	Definition
i, j	The i-th or j-the retailer, and $i, j \in N$, $i \ne j$,
N	Set of retailers $N = \{1, 2, 3, \cdots, n\}$.
Deterministic parameters:	
s_i	Maximum inventory capacity of retailer i,
k_i	Current inventory level of retailer i,
h_i	Safety stock level of retailer i,
d_i	Market demand faced by retailer i,
c_h	Inventory holding cost per unit,
c_l	Allocation price per unit,
c_b	Stockout cost per unit.
Uncertainty parameters:	
τ_i^s	Defective rate of items in the warehouse of retailer i, where $\tau_i^s \in [0, 1)$,
τ_{ij}^l	Defective rate of items allocated from retailer i to retailer j,where $\tau_{ij}^l \in [0, 1)$,
ω_i^s	Yield rate of items in the warehouse of retailer i, introduced for model construction,
ω_{ij}^l	Yield rate of products allocated from retailer i to retailer j introduced for model construction.
Decision variables:	
x_{ij}	$x_{ij} \in \{0, 1\}$, indicates whether retailer i allocates item(s) to retailer j. If $x_{ij} = 0$, it means no allocation from retailer i to retailer j, otherwise, an allocation is incurred.
l_{ij}	Quantity allocated from retailer i to retailer j.

3.4. Analysis of the UMLIA model

The processes for solving the UMLIA model established in this study considering uncertain defective rates include two main components: linearization of the proposed UMLIA model and analysis of the uncertain parameterized model.

The nonlinear structures challenge the UMLIA model; thus, a linearization preprocessing procedure is provided by employing the Big M method to equivalently transform Eq. (7). Eq. (7) is equivalently represented by Eqs. (12)–(16).

$$l_{ij} \le (1-u_{ij})*M \tag{12}$$

$$x_{ij} \le (1-u_{ij})*M \tag{13}$$

$$-x_{ij}+1 \le u_{ij}*M \tag{14}$$

$$x_{ij} \in \{0,1\}, \forall i,j \tag{15}$$

$$u_{ij} \in \{0,1\}, \forall i,j \tag{16}$$

$$i \ne j, i,j \in N$$

where u_{ij} is a newly introduced binary auxiliary variable treated as a decision variable. M is a penalty number that is theoretically assumed to be infinitely large; however, for experimental purposes, a very large constant is utilized. Given that the statistical probability-assumed uncertainty parameters are inadequate for characterizing the quantity features of defective items with small order sizes, sporadic orders, or incomplete historical demand data, the proposed uncertainty model is transformed into a robust optimization form, RMLIA. In the UMLIA model, the objective functions in Eq. (1) contains the uncertain parameters ω_j^s and ω_{ij}^l. However, because of the difficulty in handling these two uncertain parameters in the form of $(a)^+$, this study employs a new parameter OS_j to represent the cost structure of the third part. The objective function in Eq. (1) is transformed into its equivalent form as shown in Eqs. (17) and (18), respectively.

$$minf = \sum_{i\in N} c_h*\left(k_i - d_i - \sum_{j\in N} l_{ij}\right)^+ + \sum_{i\in N}\sum_{j\in N} c_l*l_{ij} + \sum_{j\in N} c_b*(OS_j)^+ \tag{17}$$

$$OS_j \ge d_j - \omega_j^s*k_j - \sum_{i\in N}\omega_{ij}^l*l_{ij} \tag{18}$$

At this moment, Eq. (18) can be furtherly transformed to an equivalently Eq. (19):

$$\sum_{i\in N}\omega_{ij}^l*l_{ij} \ge d_j - \omega_j^s*k_j - OS_j \tag{19}$$

3.5. The RMLIA model

Based on the preprocessing work, a protection function $\beta_j(l^*,\Gamma_j)$ is introduced for the allocation yield rate ω_{ij}^l (items with no defects), as shown in Eq. (20).

$$\beta_j(l^*,\Gamma_j) = max_{\{S_j\cup\{t_j\}|S_j\subseteq N_j,|S_j|=\lfloor\Gamma_j\rfloor,t_j\in N_j\setminus S_j\}}\left\{\sum_{i\in S_j}\widehat{\omega}_{ij}^l*l_{ij}^* + (\Gamma_j - \lfloor\Gamma_j\rfloor)*\widehat{\omega}_{tjj}^l*l_{ij}^*\right\} \tag{20}$$

where Γ_j is the controlling coefficient of the allocation yield rate, determined based on the risk preference of a decision-maker, with a range of $[0, |N_j|]$. N_j represents the set of coefficients ω_{ij}^l affected by parameter uncertainty for $i \in N_j$. S_j denotes a subset of N_j from which $\lfloor\Gamma_j\rfloor$ elements are selected, and the dimension of S_j is equal to the largest integer that does not exceed Γ_j, that is, $|S_j| = \lfloor\Gamma_j\rfloor$. Let $t_j := N_j \setminus S_j$ represent the remaining elements of N_j not included in S_j. It means that if the decision-maker is inclined to conservatism, a larger Γ_j is adopted to enhance the robustness of the model solution. $\overline{\omega}_{ij}^l$ and $\widehat{\omega}_{ij}^l$ represent the nominal value and the maximum deviation of the uncertain coefficient ω_{ij}^l, respectively. On the left side of Eq. (20), the protection function $\beta_j(l^*,\Gamma_j)$ is introduced. The right side of Eq. (20), by referring to the transformation method for constraints with uncertain parameters on the right side of the robust optimization models, handles the inventory yield rate ω_j^S, where $|N_j|$ denotes the dimension of N_j. The transformed Eq. (21) can be expressed as follows:

$$\sum_{i\in N_j}\overline{\omega}_{ij}^l*l_{ij} - \beta_j(l^*,\Gamma_j) \ge d_j - \left(\overline{\omega}_j^s - \frac{\Gamma_j}{|N_j|}*\widehat{\omega}_j^s\right)*k_j - OS_j \tag{21}$$

At this moment, the protection function $\beta_j(l^*,\Gamma_j)$ in Eq. (20) can be transformed into the following linear model:

$$\beta_j(l^*,\Gamma_j) = max\sum_{i\in N_j}\widehat{\omega}_{ij}^l*l_{ij}^**v_{ij} \tag{22}$$

$$s.t.\begin{cases}\sum_{i\in N_j} v_{ij} \le \Gamma_j \\ 0 \le v_{ij} \le 1, \forall i \in N_j\end{cases}$$

Based on Strong Duality theory, Eq. (22) can be transformed into its dual form, that is, Eq. (23), where z_j^a and p_{ij}^a are dual variables and are treated as decision variables during computation.

$$\beta_j(l^*,\Gamma_j) = min\sum_{i\in N_j} p_{ij}^a + \Gamma_j*z_j^a \tag{23}$$

$$s.t.\begin{cases} z_j^a + p_{ij}^a \ge \widehat{\omega}_{ij}^l*l_{ij}^*, & \forall i,j \in N \\ \sum_{i\in N_j} p_{ij}^a \ge 0, & \forall j \in N_i \\ z_j^a \ge 0, & \forall j \in N_i \end{cases}$$

Substituting Eq. (23) back into the protection function $\beta_j(l^*,\Gamma_j)$ of Eq. (21), Eq. (19) can be transformed into its robust counterpart Eq. (24) and its constraints are consistent with those in Eq. (23).

$$\sum_{i\in N_j}\overline{\omega}_{ij}^l*l_{ij} - \sum_{i\in N_j} p_{ij}^a - \Gamma_j*z_j^a \ge d_j - \left(\overline{\omega}_j^s - \frac{\Gamma_j}{|N_j|}*\widehat{\omega}_j^s\right)*k_j - OS_j \tag{24}$$

Similarly, Eq. (2) can be transformed into its robust counterpart Eq. (25).

$$\sum_{i\in N_j}\overline{\omega}_{ij}^l*l_{ij} + \sum_{i\in N_j} p_{ij}^b + \Gamma_j*z_j^b \le \left(d_j - \left(\overline{\omega}_j^s + \frac{\Gamma_j}{|N_j|}*\widehat{\omega}_j^s\right)*k_j + h_j\right)^+ \tag{25}$$

$$s.t.\begin{cases} z_j^b + p_{ij}^b \ge \widehat{\omega}_{ij}^l*l_{ij}^*, & \forall i,j \in N \\ \sum_{i\in N_j} p_{ij}^b \ge 0, & \forall j \in N_i \\ z_j^b \ge 0, & \forall j \in N_i \end{cases}$$

Based on our experience with the numerical experiments, constraint (25) leads to overly conservative decision-making results. This means that retailers prefer to accept less or no allocation to avoid exceeding safety-stock levels, thereby causing stockouts, which exactly expresses the true case in practice from the aspect of sales loss concerns for a single retail outlet. By jointly considering the situation of accepting an excessive allocation versus retailer stockouts, Eq. (25) is relaxed to Eq. (26).

$$\sum_{i\in N_j}\overline{\omega}_{ij}^{l}*l_{ij}-\sum_{i\in N_j}p_{ij}^{b}-\Gamma_j*z_j^b\le\left(d_j-\left(\overline{\omega}_j^s-\frac{\Gamma_j}{|N_j|}*\widehat{\omega}_j^s\right)*k_j+h_j\right)^+ \tag{26}$$

The constraint of Eq. (26) is consistent with Eq. (25). After the relaxation, Eq. (26) only limits the allocation quantity in the worst case to prevent retailers from experiencing excess inventory, making the decision robust for allocation recipients compared to the constraints of Eq. (25). Eq. (4) contains uncertain parameters only on the right side, specifically the inventory yield rate ω_j^s. This is processed by referring to the transformation method for constraints with uncertain parameters on the right-hand side of the robust models, Eq. (27):

$$\sum_{j\in N_i}l_{ij}\le\left(\left(\overline{\omega}_i^s-\frac{\Gamma_i}{|N_i|}*\widehat{\omega}_i^s\right)*k_i-d_i-h_i\right)^+ \tag{27}$$

In summary, the UMLIA considering uncertain defective rates is reformulated as the RMLIA with the objective function in Eq. (17) and constraints in Eqs. (5) and (6) and (8)–(16) and (24)–(27). The complete RMLIA is reformulated as follows: Eqs. (29), (30) and (32) are derived from Eqs. (2), (4) and (18), respectively, whereas Eqs. (31)–(43) correspond to Eqs. (3)–(16) (excluding Eq. (4)).

$$minf=\sum_{i\in N}c_h*\left(k_i-d_i-\sum_{j\in N}l_{ij}\right)^++\sum_{i\in N}\sum_{j\in N}c_l*l_{ij}+\sum_{j\in N}c_b*(OS_j)^+ \tag{28}$$

$$\sum_{i\in N_j}\overline{\omega}_{ij}^{l}*l_{ij}-\sum_{i\in N_j}p_{ij}^{a}-\Gamma_j*z_j^a\ge d_j-\left(\overline{\omega}_j^s-\frac{\Gamma_j}{|N_j|}*\widehat{\omega}_j^s\right)*k_j-OS_j \tag{29}$$

$$s.t.\begin{cases}z_j^a+p_{ij}^a\ge\widehat{\omega}_{ij}^{l}*l_{ij}^*, & \forall i,j\in N\\ \sum_{i\in N_j}p_{ij}^a\ge 0, & \forall j\in N_i\\ z_j^a\ge 0, & \forall j\in N_i\end{cases}$$

$$\sum_{i\in N_j}\overline{\omega}_{ij}^{l}*l_{ij}-\sum_{i\in N_j}p_{ij}^{b}-\Gamma_j*z_j^b\le\left(d_j-\left(\overline{\omega}_j^s-\frac{\Gamma_j}{|N_j|}*\widehat{\omega}_j^s\right)*k_j+h_j\right)^+ \tag{30}$$

$$s.t.\begin{cases}z_j^b+p_{ij}^b\ge\widehat{\omega}_{ij}^{l}*l_{ij}^*, & \forall i,j\in N\\ \sum_{i\in N_j}p_{ij}^b\ge 0, & \forall j\in N_i\\ z_j^b\ge 0, & \forall j\in N_i\end{cases}$$

$$\sum_{i\in N}l_{ij}+k_j\le s_j,\forall i\ne j,j\in N \tag{31}$$

$$\sum_{j\in N_i}l_{ij}\le\left(\left(\overline{\omega}_i^s-\frac{\Gamma_i}{|N_i|}*\widehat{\omega}_i^s\right)*k_i-d_i-h_i\right)^+ \tag{32}$$

$$\omega_{ij}^{l}=1-\tau_{ij}^{l}, \tag{33}$$

$$\omega_i^s=1-\tau_i^s, \tag{34}$$

$$l_{ij}\le(1-u_{ij})*M \tag{35}$$

$$x_{ij}\le(1-u_{ij})*M \tag{36}$$

$$-x_{ij}+1\le u_{ij}*M \tag{37}$$

$$x_{ij}\in\{0,1\},\forall i,j \tag{38}$$

$$u_{ij}\in\{0,1\},\forall i,j \tag{39}$$

$$i\ne j,i,j\in N$$

$$\sum_{i\in N}x_{ij}\le n-1,\forall i\ne j,j\in N \tag{40}$$

$$\sum_{j\in N}x_{ij}\le 1,\forall i\ne j,i\in N \tag{41}$$

$$0\le x_{ij}+x_{ji}\le 1,\forall i\ne j,i,j\in N \tag{42}$$

$$s_i,k_i,d_i,l_{ij}\ge 0,\forall i,j\in N \tag{43}$$

4. Solving method

For a robust optimization model, decision-makers must address two challenges: ascertaining the extent of uncertainty of an uncertain parameter and justifying the performance of the RMLIA decision. Both these concerns are discussed in the following subsections.

4.1. Processing of the uncertainty parameter

The pivotal variable method is introduced to determine the value of the uncertainty parameter. Specifically, a supplier selects m items from a sample and records whether an item is free of defects. If the stochastic probability of each item being qualified or defective is independent and identically distributed, then the sampling process exhibits the characteristics of a Bernoulli trial. Let x_i^c represent the quality of the item. If the i-th item is qualified, then x_i^c is recorded as 1; otherwise, x_i^c is recorded as 0. In other words, x_i^c is expressed by Eq. (44):

$$x_i^c=\begin{cases}1 & \text{item with quality defect}\\ 0 & \text{item without quality defect}\end{cases} \tag{44}$$

If X^c denotes the number of qualified items in the sample, then $X^c=\sum_{i=1}^m x_i^c$. Here, p and $\widehat{p}$ are temporarily denoted as the means of the population and sample of items with no defects, respectively. It can be observed that $\widehat{p}$ can be expressed as $\widehat{p}=X^c/m$ and applied to approach p. As the probability of each item being a qualified or defective item is independent and identically distributed, x_i^c follows a Bernoulli distribution, namely, $x_i^c\sim B(1,p)$, and the variance of x_i^c is $D(x_i^c)=p*(1-p)$. The standard deviation S of x_i^c is approximately.

$$S=\sqrt{\frac{D(x_i^c)}{m}}\approx\sqrt{\frac{\widehat{p}*(1-\widehat{p})}{m}} \tag{45}$$

The pivotal variable method is shared to determine the confidence interval for the qualification rate, where the confidence interval P_0 is defined as P_0: {Estimate ± Standard error * Critical value}, where "Estimate" is the sample mean, "Standard error" is the sample standard deviation, and the "critical value" is the z-score. If x_i^c does not follow a specific distribution over the population, based on the Central Limit theorem, when the sample size m is sufficiently large, the sampling distribution reflecting the mean and variance of the population approaches a normal distribution. Therefore, the z-score is adopted based on an approximation of the normal distribution $(0,1)$.

For example, if the confidence level is 68.2%, the confidence interval should be $[\mu-\sigma,\mu+\sigma]$. This implies that the z-score corresponds to the critical value probability $P(Z<(1+C)/2)$, as shown in Fig. 2. By

checking the critical value table of the standard normal distribution, the z-score is equal to 1. Based on this assumption, the nominal value and maximum deviation in the robust model in this study can be ascertained under different confidence preferences, which can also be employed to evaluate decision effectiveness by comparing the robust model with that of a stochastic model.

Determination of the confidence interval should adhere to two principles: (1) maximizing the confidence level and (2) minimizing the range of the confidence interval. In practice, suppliers have already conducted surveys to estimate the approximate defective rate, although it should be noted that the actual defective rate is often much higher than the defective rate after production. Therefore, the actual choice of the z-score is based on the decision-maker's own risk preference and experience. For ease of analysis, this study chooses 2 as the z-score reference value for the numerical experiments, as shown in Table 3, which lists the nominal and maximum deviation values of the robust uncertainty parameters at different confidence levels.

Table 3
Nominal values and maximum deviations of uncertainty parameters at different confidence levels.

z-score	Confidence level $C(\%)$	Confidence interval P_0	Nominal value $\overline{\omega}$	Maximum deviation value $\widehat{\omega}$
0.5	38.29	$\widehat{p} \pm 0.5^*\sqrt{\frac{\widehat{p}^*(1-\widehat{p})}{m}}$	$\widehat{p}$	$0.5^*\sqrt{\frac{\widehat{p}^*(1-\widehat{p})}{m}}$
1.0	68.27	$\widehat{p} \pm 1.0^*\sqrt{\frac{\widehat{p}^*(1-\widehat{p})}{m}}$	$\widehat{p}$	$1.0^*\sqrt{\frac{\widehat{p}^*(1-\widehat{p})}{m}}$
1.5	86.64	$\widehat{p} \pm 1.5^*\sqrt{\frac{\widehat{p}^*(1-\widehat{p})}{m}}$	$\widehat{p}$	$1.5^*\sqrt{\frac{\widehat{p}^*(1-\widehat{p})}{m}}$
2.0	95.45	$\widehat{p} \pm 2.0^*\sqrt{\frac{\widehat{p}^*(1-\widehat{p})}{m}}$	$\widehat{p}$	$2.0^*\sqrt{\frac{\widehat{p}^*(1-\widehat{p})}{m}}$
2.5	98.76	$\widehat{p} \pm 2.5^*\sqrt{\frac{\widehat{p}^*(1-\widehat{p})}{m}}$	$\widehat{p}$	$2.5^*\sqrt{\frac{\widehat{p}^*(1-\widehat{p})}{m}}$
3.0	99.73	$\widehat{p} \pm 3.0^*\sqrt{\frac{\widehat{p}^*(1-\widehat{p})}{m}}$	$\widehat{p}$	$3.0^*\sqrt{\frac{\widehat{p}^*(1-\widehat{p})}{m}}$
…	…	…	$\widehat{p}$	…

4.2. Design of a performance evaluation metric

To verify the feasibility of the robust parameters and the performance of the robust decision, a comparative experiment is designed based on the pivot-variable method. Specifically, the comparative experiment contrasts the values of the uncertainty parameters used in this study with those derived from traditional methods of handling stochastic defective rates (the expectation method) to analyze the optimization effectiveness. The process of constructing the performance evaluation metric for the comparative experiment on the UMLIA and RMLIA is illustrated in Fig. 3.

According to Fig. 3, the sample average approximation method and robust optimization method both need to know the mean defective rate $\widehat{p}$ in advance, but the former sampling method requires more information on the uncertain defective rate, whereas the latter method only requires the mean value and z-score, which are easy to obtain from decision-makers. In the sampling average approximation method, the random defective rate follows a specific distribution with a mean of $\widehat{p}$. In the numerical experiments, a number within the range $[0, 1]$ is generated based on this distribution. The difference (RC), between the uncertainty model and the robust model is then recorded, which is expressed by Eq. (46). The same processes are repeated at a preset number of iterations, and the recorded cost differences are accumulated as performance evaluation metrics.

$$RC = \sum_{i\in N} c_h{}^*\left(k_i - d_i - \sum_{j\in N} l_{ij} + \sum_{j\in N} l_{ji}\right)^+ + \sum_{i\in N}\sum_{j\in N} c_l{}^* l_{ij} + \sum_{j\in N} c_b{}^*\left(d_j - \omega_j^s{}^* k_j - \sum_{i\in N} \omega_{ij}^l{}^* l_{ij} + \sum_{i\in N} \omega_{ji}^l{}^* l_{ji}\right)^+ \tag{46}$$

In Eq. (46), ω_j^s is the qualification rate generated from the assumed distribution, and ω_{ij}^l represents the robust qualification rate computed based on the pivotal variable method. By drawing a curve representing the changing trend of the cost difference with the obtained RCs, we can obtain more information from the experiments; if a steadily increasing accumulated cost difference is observed, it indicates that, in most cases, the cost of the robust model is lower than that of the sampling average approximation method. Therefore, this study uses this performance, indicating a curve, to justify the performance of the traditional and robust methods.

5. Numerical experiment

The numerical experiments in this study are conducted on a Windows OS with 16.0 GB RAM. The model is solved using the IBM ILOG CPLEX Optimization Studio (version 12.10.0.0). MATLAB R2019a is used to compose the comparative tables and figures. The experiments use an inventory allocation system comprising 10 retailers for illustration purposes. Table 4 presents the inventory information for each retailer.

The inventory holding cost per unit c_h for the items in Table 4 is 2 Yuan/item. The allocation price c_l is 3 Yuan/item, and the stockout cost c_b is 20 Yuan/item. We assumed that soon after the supplier produced a batch of items, a sample of 1,000 items was taken for measurement, and the defective rate for this sample is 0.1.

5.1. Comparation and analysis of the computation results

The experiment is designed to analyze a scenario in which the defective rate is equal to the mean of the sample data (a uniform distribution is assumed). The actual qualification rate of the items (calculated based on the random defective rate) is assumed to follow a uniform distribution $U(0.84, 0.96)$. In Table 5, the row labels represent the retailer numbers from the supplying side, and the column labels denote

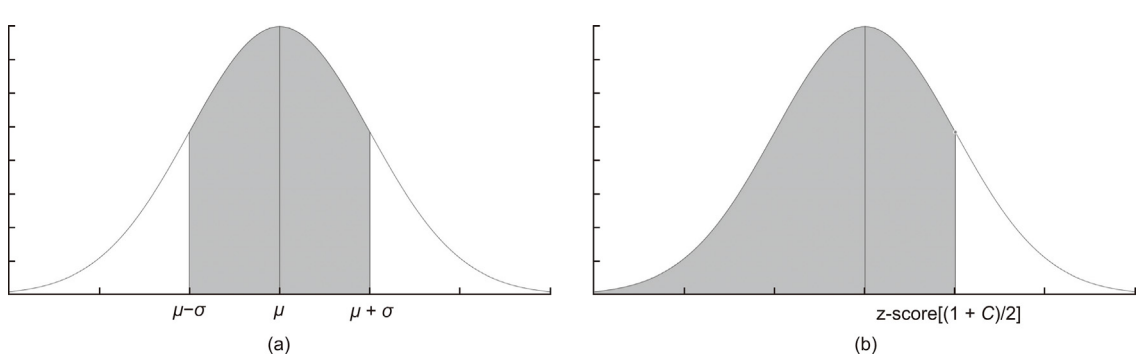

Fig. 2. An illustration of (a) confidence interval under the confidence level of 68.2% and (b) z-score under the confidence level 68.2%.

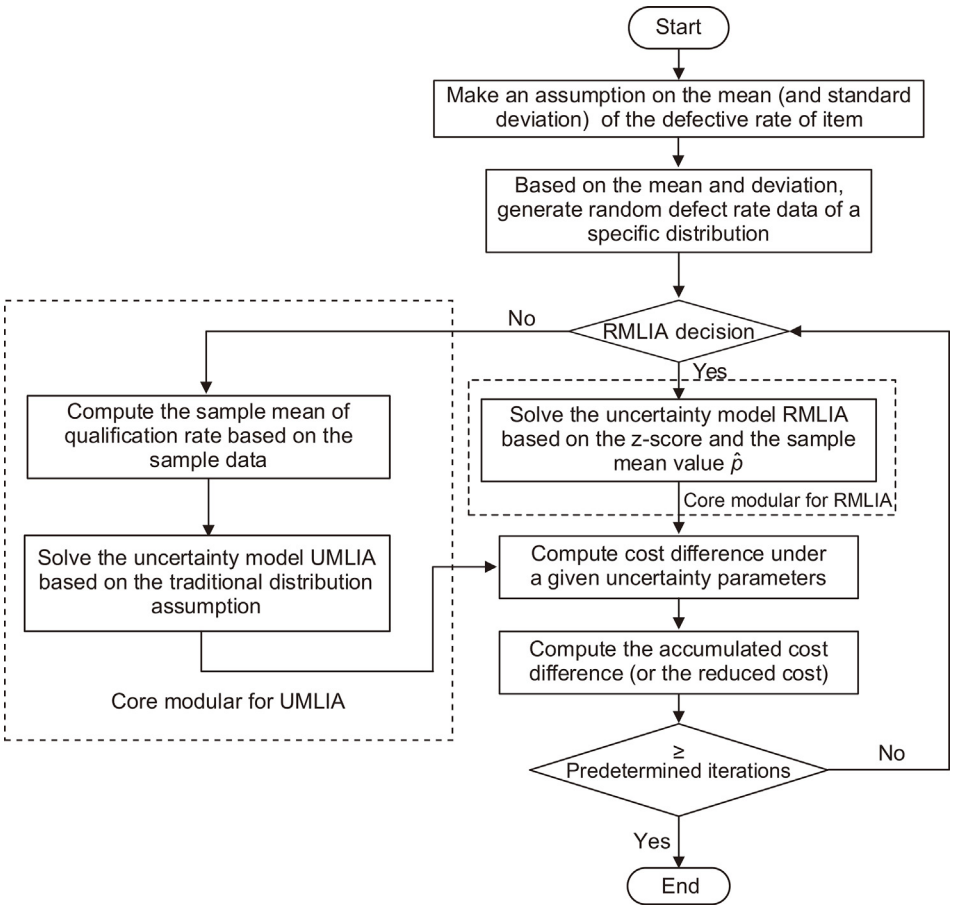

Fig. 3. Calculation flow under the performance evaluation metric. UMLIA and RMLIA.

the retailer numbers from the recipient side. For the sampling average approximation method, decisions are made by substituting the defective rate with the random data under the sample mean of 0.1 in the UMLIA model, and the decision results are shown as the numbers outside the brackets in Table 5. For RMLIA, the decision is made based on a nominal value of 0.1 and $\Gamma_j = 2$. The decision results of the RMLIA are shown as numbers inside the brackets in Table 5.

Table 5 illustrates a typical phenomenon in item allocation; the allocation result is sparse for the entire system. The decision results from the sampling average approximation method are as follows: Retailer 6 and Retailer 10 allocate approximately 9.4×10^4 items and 40,444 items, respectively, to Retailer 4; Retailer 2 allocates approximately 47,778 items to Retailer 5; and Retailer 3 allocates 1.2×10^4 items to Retailer 8. Zeros indicate that there are no allocation operations. In contrast, the decision results from the robust optimization model are as follows: Retailers 6 and 10 allocate 93,392 and 41,665 items, respectively, to Retailer 4; Retailer 2 allocates 48,108 items to Retailer 5; and Retailer 3 allocates 11,696 items to Retailer 8.

Table 4
Inventory information of 10 retailers.

Retailer i	Maximum stock s_i	Current inventory level k_i	Safety stock point h_i	Market demand d_i
1	18	5	1	4
2	25	16	3	5
3	30	8	3	3
4	20	1	3	13
5	25	3	3	7
6	20	16	2	3
7	22	4	3	5
8	25	5	3	6
9	20	4	2	4
10	16	12	1	3

To further analyze the impact of different uncertainty extent of Γ_j on the decision results, this paper supplements the experiment by setting Γ_j to 5 (medium robustness) and 8 (high robustness). The decision results from the two methods are obtained under qualification rates of following $U(0.84, 0.96)$, $U(0.82, 0.98)$, and $U(0.80, 1.00)$, respectively. The sampling frequency is 10,000. Box plots and performance-indicating curves are used to illustrate the results. The computational results for the different distributions are shown in Figs. 4 and 5.

Fig. 4 shows that, under three groups of distribution ranges of the defective rate, that is, $U(0.84,0.96)$, $U(0.82,0.98)$, and $U(0.80,1.00)$, the total cost obtained through the sampling average approximation method and robust optimization method under the three robustness levels exhibit similar distributions but with different magnitudes. When the certainty deviation reaches its maximum ($\Gamma_j = 8$), the robust model presents the minimum total cost under all the distribution ranges of defective rates. Furthermore, as the distribution range of the defective rate uncertainty increase, the total cost also increases. This implies that the larger the uncertainties considered, the greater the system cost incurred.

Fig. 5 reveals that, under three groups of distribution ranges of the defective rate, that is, $U(0.84,0.96)$, $U(0.82,0.98)$, and $U(0.80,1.00)$, the total costs obtained through the sampling average approximation method and robust optimization method under three robustness levels exhibit an overall upward trend except for the situation under $U(0.84, 0.96)$ and $\Gamma_j = 8$. It indicates that for the situation with more certain information, the risk-aggressive ($\Gamma_j = 8$) decision shows weakness in the decision results, the robust optimization method can obtain better results in higher uncertainty cases. This also indicates that as the defective rate becomes more dispersed, the robust optimization method

Table 5
Cumulative cost difference (*RC*) of allocation decision systems (10^4).

Retailer *i.j*	1	2	3	4	5	6	7	8	9	10
1	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
2	0 (0)	0 (0)	0 (0)	0 (0)	4.7778 (4.8108)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
3	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (1.1696)	1.2 (0)	0 (0)	0 (0)
4	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
5	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
6	0 (0)	0 (0)	0 (0)	9.4 (9.3392)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
7	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
8	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
9	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
10	0 (0)	0 (0)	0 (0)	4.0044 (4.1665)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)

becomes superior to the traditional sampling average approximation method. This demonstrates that the robust model is more resilient to deviations between the actual and sampled defective rates (i.e., the degree of defective rate dispersion).

Table 6 presents the computational costs (mean and variance) of each model for 10,000 experiments, supporting the conclusions shown in Fig. 5. The robust optimization method consistently yields lower total costs, except in the scenario where the defective rate has low dispersion. However, $\Gamma_j = 8$, in such case, the overly conservative parameters result in higher total costs compared to the sampling (average approximation) method. The large-scale experimental results demonstrate that as the degree of uncertainty increases, the performance advantage of the robust optimization method becomes more pronounced.

5.2. Sensitive analysis on the allocation effects

In practice, reducing stockout losses is the primary purpose of allocation strategies because stockouts not only reduce retailers' profits but also lead to customer dissatisfaction. In particular, during the era of mobile internet development, stockouts in e-business exhibit a potential amplifying effect (Jiang et al., 2021). Therefore, this section analyzes the effectiveness of allocation strategies in mitigating stockout costs while accounting for defective items. To ensure timely order fulfillment, stockout costs are typically incorporated into pricing strategies to sustain service-level agreements. The experiment assumes that the qualification rate of the items follows a uniform distribution $U(0.82, 0.98)$, with unit stockout costs c_b of 20, 25, and 30 Yuan per item. Under these two cost scenarios, *RC*s between the sampling method and the robust optimization method (with low, medium, and high Γ_j parameters) are evaluated through 10,000 experiments.

Based on 10,000 experiments under U (0.82,0.98), Table 7 compares the total cost results (mean and variance) of UMLIA obtained by the sampling method and RMLIA obtained by the robust optimization, as well as *RC*s under three levels of unit shortage costs $c_b = 20$, 25, and 30 for $\Gamma_j = 2$ (low robustness), 5 (medium robustness), and 8 (high robustness), respectively. Table 8 further validates the findings shown in Table 7 for $U(0.80, 1.00)$.

Tables 7 and 8 indicate that, as the extent of uncertainty increases, the robust optimization decision demonstrates greater advantages in reducing shortage costs. Furthermore, it reveals that a larger c_b would cause more shortage costs for the system, and greater uncertainty would result in higher costs. The combined effects of c_b and uncertainty have serious impacts on system shortages; thus, for decision-makers, it is necessary to monitor the quality of items, especially for items with high prices or shortage costs.

To clearly illustrate the evolving processes of *RC*, the performance-indicating curves are shown in Fig. 6.

Fig. 6 shows that under medium-to-high unit stockout costs and medium-to-high levels of defective rate dispersion, the three performance indicator curves exhibit an overall upward trend. Furthermore, as the stockout costs increase, the cumulative cost differences become more pronounced. This indicates that, in scenarios with higher unit stockout costs, the decision outcomes of the robust method are superior to those of the traditional method.

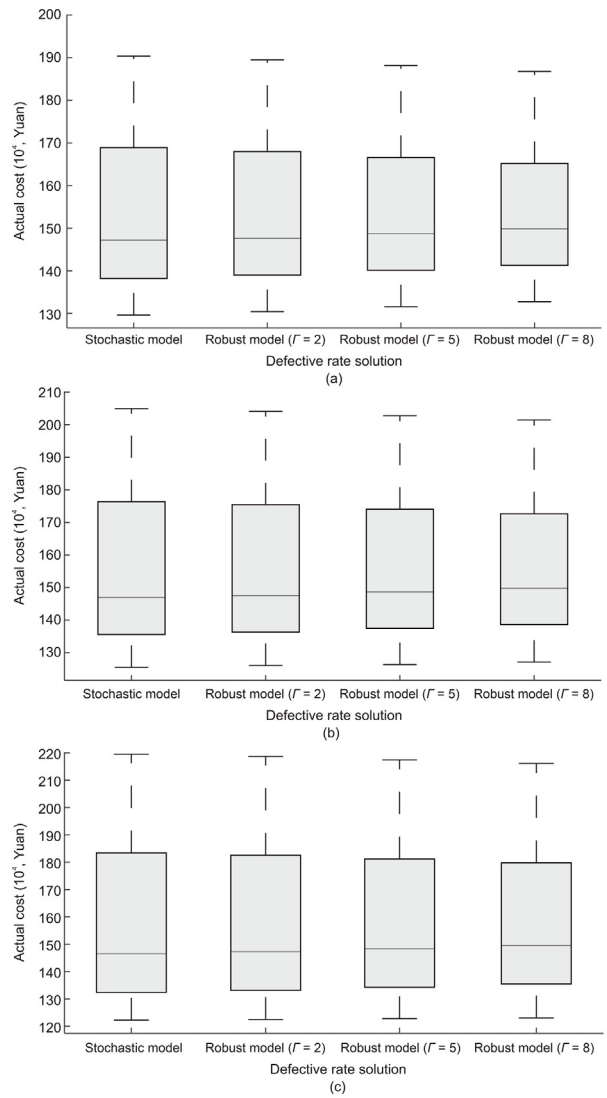

Fig. 4. Box plot of objective result distribution of UMLIA and RMLIA. (a) Total cost distribution under $U(0.84, 0.96)$; (b) total cost distribution under $U(0.82, 0.98)$; (c) total cost distribution under $U(0.80, 1.00)$.

6. Discussion and implications

Currently, certain theoretical studies often neglect the intrinsic uncertainty characteristics of defective rates but involve many decision-

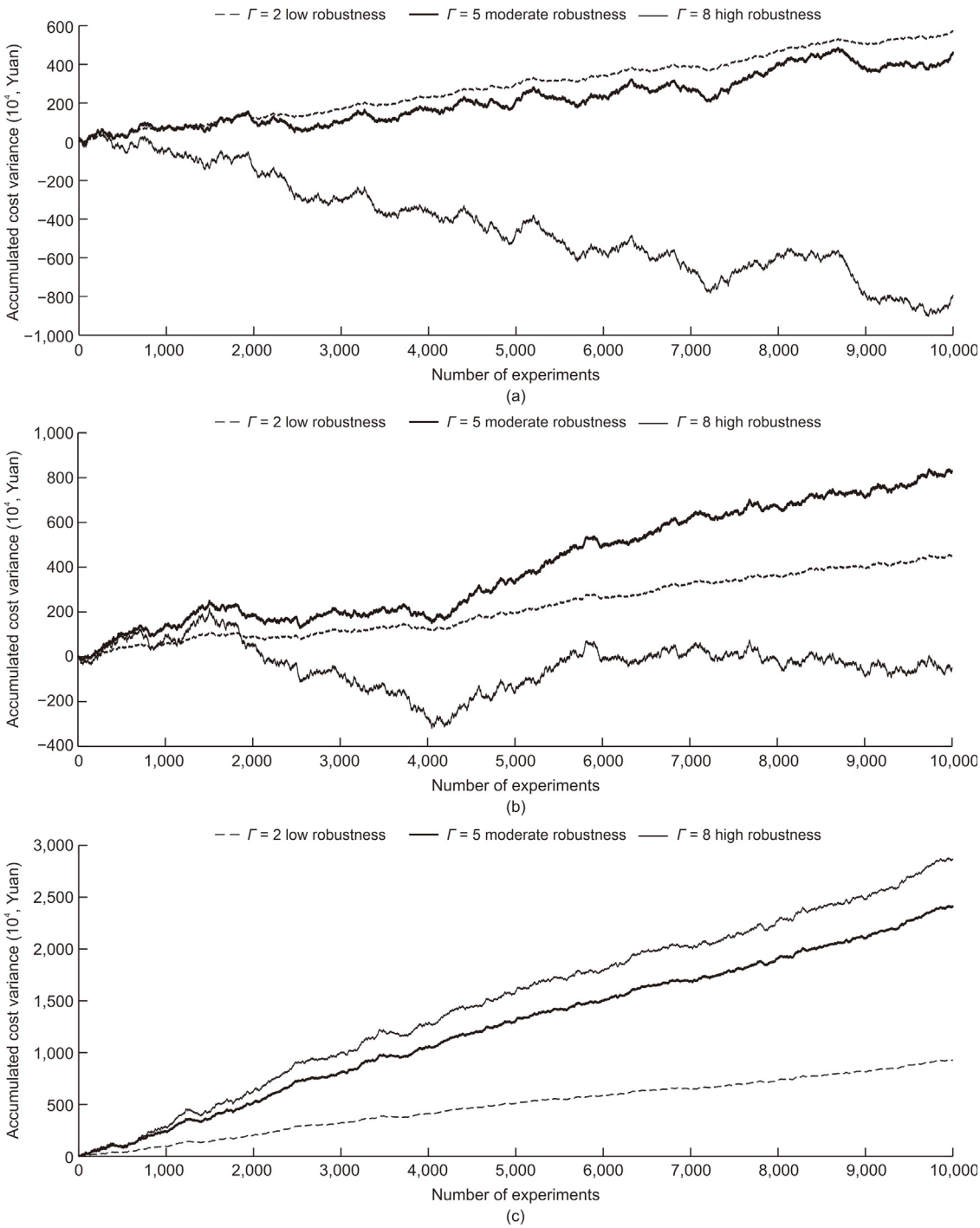

Fig. 5. Performance indicating curves under three level of uncertainties. (a) The defective rate under $U(0.84, 0.96)$; (b) the defective rate under $U(0.82, 0.98)$; (c) the defective rate under $U(0.80, 1.00)$.

Table 6
Cumulative cost differences (RC) of allocation decision systems based on sampling and robust methods (under different levels of defective rates Γ_j).

Defective rate U	Sampling methods (mean, variance)	Robust methods (mean, variance)			RC (10^4 Yuan)		
		$\Gamma_j = 2$	$\Gamma_j = 5$	$\Gamma_j = 8$	$\Gamma_j = 2$	$\Gamma_j = 5$	$\Gamma_j = 8$
$U(0.84, 0.96)$	153.34, 322.14	153.29, 297.25	153.31, 261.37	153.44, 227.65	478.81	306.93	−1,006.4
$U(0.82, 0.98)$	155.56, 569.54	155.50, 536.92	155.45, 491.45	155.52, 445.18	554.81	1,123.71	422.27
$U(0.80, 1.00)$	158.08, 879.39	157.99, 841.58	157.84, 789.73	157.80, 736.39	880.7	2,368.73	2,794.18

Table 7
Difference (RC) of allocation results based on sampling and robust optimization methods ($U(0.82, 0.98)$).

c_b	Sampling methods (mean, variance)	Robust optimization methods (mean, variance)			RC (10^4 Yuan)		
		$\Gamma_j = 2$	$\Gamma_j = 5$	$\Gamma_j = 8$	$\Gamma_j = 2$	$\Gamma_j = 5$	$\Gamma_j = 8$
20	155.56, 569.54	155.50, 536.92	155.45, 491.45	155.52, 445.18	554.81	1,123.71	422.27
25	169.28, 887.52	169.15, 836.51	168.99, 765.18	168.976, 692.82	1,279.76	2,882.19	3,003.08
30	182.03, 1,273.92	181.85, 1,200.60	181.63, 1098.14	181.59, 994.18	1,793.46	4,019.70	4,399.54

Table 8
Difference (RC) of allocation results based on sampling and robust methods ($U(0.80, 1.00)$)

c_b	Sampling methods (mean, variance)	Robust methods (mean, variance)			RC (10^4 Yuan)		
		$\Gamma_j = 2$	$\Gamma_j = 5$	$\Gamma_j = 8$	$\Gamma_j = 2$	$\Gamma_j = 5$	$\Gamma_j = 8$
20	158.08, 879.39	157.99, 841.58	157.84, 789.73	157.80, 736.39	880.70	2,368.73	2,794.18
25	172.60, 1375.96	172.43, 1316.77	172.16, 1,235.36	172.01, 1,151.67	1,724.75	4,491.45	5,989.79
30	186.67, 1981.38	186.43, 1,896.15	186.04, 1,778.91	185.80, 1,658.40	2,453.10	6,356.35	8,748.95

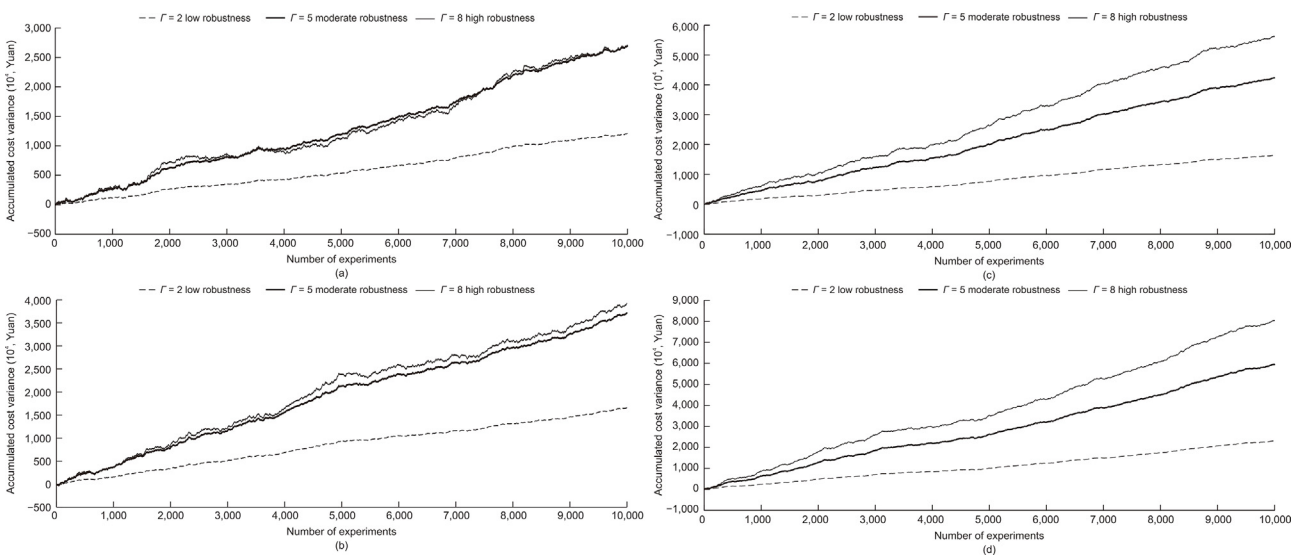

Fig. 6. Performance indicating curves of RC. (a) $U(0.82, 0.98)$ and $c_b = 25$; (b) $U(0.82, 0.98)$ and $c_b = 30$; (c) $U(0.80, 1.00)$ and $c_b = 25$; (d) $U(0.80, 1.00)$ and$c_b = 25$.

makers' subjective risk preferences and judgements, resulting in the decision results being less adaptive and robust in real scenarios. In this study, in the context of a multi-location retail outlet network with limited sales volume in each location, a robust optimization model based on budget uncertainty sets is developed, replacing idealized assumptions on defective rates with a fixed proportion or following presupposed probability distributions on scaling logistics operations. Furthermore, a quantitative association mechanism between risk preferences and uncertainty parameter intervals is established using the pivotal variable method. This theoretical extension demonstrates that in high-volatility retail sectors such as the fast fashion industry, the unpredictability of defective rates should be modeled as an endogenous decision variable rather than treated as exogenous noise.

At the theoretical level, unlike prior studies that rely solely on probabilistic assumptions about general products or commodities, this study captures the fluctuation range of defective rates via budget uncertainty sets for limited sales volume products, while enabling decision-makers to adjust parameter confidence intervals based on their risk tolerance. The constructed mathematical model can be readily extended to real-world decision-making contexts, including those involving luxury or branded apparel, automotive sales, and the maintenance of heavy equipment. These sectors often operate with small or limited storage capacities, which can lead to abnormal allocation requirements. Furthermore, as an effective evaluation tool, the proposed cumulative cost difference curve enables a visual comparison of cost deviations across different models under varying defective rates. This approach is particularly suitable for optimizing uncertainty problems that require balancing risks and costs.

From a managerial perspective, this study's findings provide theoretical and implementation support for allocation practices among multi-location retailers with small sales volumes. Based on our study, we can derive managerial insights from three areas. First, the efficacy of the proposed robust optimization model has been confirmed through numerical experiments, demonstrating its ability to significantly reduce emergency allocation costs and stockout losses stemming from elevated defective rates. It is particularly recommended for deployment in scenarios involving products with high volatility and demand as well as for items that undergo rapid technique iterations, where such a model can provide substantial benefits in managing uncertainty and optimizing inventory performance. Second, particular emphasis should be placed on establishing cross-departmental data-sharing platforms that integrate warehouse quality inspection data with store return or repair records because the estimation of defective rates has been identified as susceptible to subjective judgmental biases. Historical data analysis should be systematically employed to calibrate the uncertainty parameter intervals for optimization models, where machine learning or advanced artificial intelligence can be considered jointly with expert decision-making. Finally, sensitivity analysis reveals a critical correlation between

stockout costs and decision robustness, providing empirical support for differential inventory policies. For stores with high stockout costs or risks, conservative inventory buffers must be implemented to ensure robustness. Conversely, tolerance of higher defective rates is accepted in low-stockout-risk regions to reduce inventory-holding costs.

7. Conclusion

This study investigated multi-location allocation decisions in the context of uncertain defective items. A robust allocation model was constructed that focused on minimizing the total cost of the allocation system. In ascertaining the uncertain parameters, the pivotal variable method was referenced to determine the range of uncertainty parameters for the defective rate. In addition, considering the challenges of evaluating the decision effectiveness of the robust optimization model, especially how to justify the robust optimization decision effects from a large-scale statistical perspective, this study developed a new metric by examining the trending curve formed from the cumulative total cost differences between the UMLIA model obtained by the sampling average approximation model and the RMLIA yielded from the sample-data-based robust optimization model.

The results of the numerical experiments indicate the superiority of the robust optimization decision over the sampling-based decision in managing stockouts by reducing the shortage cost. Moreover, because less information on the true distribution of data is required by robust optimization decisions, it is evident that robust optimization decisions have strong potential for solving problems involving uncertainty factors, particularly for environments with relatively high disturbances of uncertain parameters.

In future research, the multi-location allocation problem considering occasional temporary allocation requirements will be extended to consider more uncertainty factors, such as defective rates of multiple items and demand uncertainty, to investigate uncertain demand pre-allocation problems. Correspondingly, model-feature-based solving approaches have been planned.

CRediT authorship contribution statement

Ligang Cui: Writing – original draft, Visualization, Supervision, Software, Funding acquisition, Conceptualization. **Zuguo Peng:** Writing – original draft, Visualization, Data curation. **Yingcong Chen:** Writing – original draft, Visualization, Validation, Software, Investigation. **Wenjing Xie:** Writing – review & editing, Data curation.

Declaration of competing interest

The authors declare that there are no conflicts of interest.

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